

Topic 5

Geometry

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions**

Learning focus

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

5.1	Geometrical language and angle facts
5.2	Polygons and symmetry
5.3	Congruence and similarity
5.4	Circle theorems
5.5	Constructions, loci and bearings
5.6	Investigating geometric patterns

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Angles and polygons

Angles on a straight line total 180° , around a point 360° and inside a triangle 180° . Parallel lines yield corresponding and alternate equal angles and co-interior supplementary angles. A polygon with n sides has interior angle sum $(n - 2) \times 180^\circ$.

Worked example

A regular octagon has interior angle $(8 - 2) \times 180^\circ / 8 = 135^\circ$.

Exam check: Name the angle rule in a reasoning question; a bare numerical value may not show why.

2. Congruence and similarity

Congruent triangles match exactly in side lengths and angles, often by SSS, SAS, ASA or RHS. Similar figures have equal corresponding angles and lengths in a common ratio k . Corresponding areas change by k^2 and volumes by k^3 . Match corresponding vertices before calculating.

Worked example

Similar triangles have sides 4 cm and 10 cm corresponding; $k = 2.5$. If small area is 12 cm^2 , larger area $= 12 \times 2.5^2 = 75 \text{ cm}^2$.

Exam check: A length scale factor must be squared for areas.

3. Circle geometry

An angle at the centre is twice the angle at the circumference on the same arc. Angles in the same segment are equal; the angle in a semicircle is 90° . Opposite angles in a cyclic quadrilateral sum to 180° . The radius is perpendicular to the tangent at contact.

Worked example

If a centre angle subtending an arc is 116° , the corresponding circumference angle is 58° .

Exam check: Make sure the angles stand on the same arc; a sketch prevents using the wrong theorem.

4. Constructions and loci

A perpendicular bisector is the locus equidistant from two fixed points. An angle bisector is the locus equidistant from the sides of an angle. The locus at a fixed distance from a point is a circle; from a line it is two parallels plus semicircular end caps for a finite segment. Show arcs in a compass construction.

Worked example

Points 3 cm from A form a circle of radius 3 cm centred at A. Intersect it with a perpendicular bisector to satisfy both conditions.

Exam check: Read whether the question asks for a shaded region, a boundary or individual intersection points.

5. Bearings and transformations of shape

Bearings are measured clockwise from north and written with three digits. In a scale diagram, draw north lines at both points; reverse bearings differ by 180° modulo 360° . Reflections preserve size but reverse orientation; rotations preserve size and orientation.

Worked example

The bearing of B from A is 065° . The bearing of A from B is 245° .

Exam check: Always measure a bearing from north at the starting point of that journey.

6. Investigating geometric patterns

Measure several examples, record angle or length data systematically and make a conjecture. A conjecture based on diagrams needs justification from geometric rules. Try an irregular counterexample to test whether a rule relies on regularity.

Worked example

Regular polygons with 4,5,6 sides have exterior angles $90^\circ, 72^\circ, 60^\circ$. The general rule is $360^\circ/n$.

Exam check: Measurements can be imprecise; use angle sums to justify the formula.

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Graphical calculator and Paper 6

A dynamic geometry sketch can suggest a theorem. Give the angle reason or a counterexample; a screen image alone is not proof.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.

Practice paper

Topic 5 · Geometry

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:	
Class / date:	

Total: 36 marks

Section A · Multiple choice

Choose one answer for each question.

1. Sum of angles in a triangle? [1]

- A 90°
- B 180°
- C 270°
- D 360°

2. Interior angle sum of a hexagon? [1]

- A 360°
- B 540°
- C 720°
- D 900°

3. One exterior angle of a regular decagon? [1]

- A 18°
- B 36°
- C 72°
- D 144°

4. If lengths scale by 3, areas scale by [1]

- A 3
- B 6
- C 9
- D 27

5. A diameter subtends what angle at the circumference? [1]

- A 45°
- B 90°
- C 180°
- D 360°

6. The radius meets a tangent at [1]

- A 30°
- B 45°
- C 60°
- D 90°

7. The perpendicular bisector is equidistant from [1]

- A two fixed points
- B two intersecting lines
- C a line and a point
- D the sides of an angle

8. Reverse bearing of 045° ? [1]

- A 135°
- B 180°
- C 225°
- D 315°

Section B · Structured questions

9. A regular polygon has an exterior angle of 24° .

(a) Find its number of sides. [2]

(b) Find one interior angle. [1]

(c) Find the sum of its interior angles. [2]

[Total: 5]

10. Two similar triangles have corresponding sides of 6 cm and 15 cm. The smaller triangle has area 18 cm^2 .

(a) Calculate the length scale factor. [1]

(b) Calculate the area scale factor. [2]

(c) Find the larger triangle area.

[2]

[Total: 5]

11. In a circle, the centre angle AOB is 124° and C is on the major arc AB.

(a) Find angle ACB.

[2]

(b) State the theorem used.

[1]

(c) Find angle OAB in the isosceles triangle AOB.

[2]

[Total: 5]

12. The bearing of Q from P is 118° .

(a) Find the bearing of P from Q.

[2]

(b) Find the angle clockwise from north to a due west direction.

[1]

(c) Explain why a bearing must be measured at the point of departure.

[2]

[Total: 5]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. An investigation records one exterior angle of regular polygons with $n=4,5,6$ sides.

(a) Calculate the three exterior angles. [2]

(b) Suggest a formula $e(n)$ in degrees. [2]

(c) Use it to predict the angle for $n=12$. [2]

(d) Explain why the rule applies only to regular polygons here. [2]

[Total: 8]

Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. B — 180° . All interior angles of any triangle total 180° .
2. C — 720° . $(6-2) \times 180^\circ = 720^\circ$.
3. B — 36° . $360^\circ/10 = 36^\circ$ for one exterior angle.
4. C — 9. Area changes by the square of the length factor: $3^2 = 9$.
5. B — 90° . The angle in a semicircle is a right angle.
6. D — 90° . The tangent is perpendicular to the radius at contact.
7. A — two fixed points. Every point on a perpendicular bisector is equidistant from the endpoints.
8. C — 225° . The reverse bearing is $045^\circ + 180^\circ = 225^\circ$.

Section B · Worked solutions

9. a) $360^\circ/24^\circ = 15$ sides [2]. b) $180^\circ - 24^\circ = 156^\circ$ [1]. c) $(15 - 2) \times 180^\circ = 2340^\circ$ [2].

10. a) $k = 15/6 = 2.5$ [1]. b) $k^2 = 6.25$ [2]. c) $18 \times 6.25 = 112.5 \text{ cm}^2$ [2].

11. a) Angle $ACB = 124^\circ/2 = 62^\circ$ [2]. b) Angle at centre is twice angle at circumference on the same arc [1]. c) $OA = OB$; each base angle $= (180^\circ - 124^\circ)/2 = 28^\circ$ [2].

12. a) $118^\circ + 180^\circ = 298^\circ$ [2]. b) 270° [1]. c) A bearing is the clockwise direction from north at its starting point; changing the starting point changes the reference north line [2].

Section C · Investigation and modelling solutions

13(a). $90^\circ, 72^\circ, 60^\circ$ [2].

13(b). Equal exterior angles sum to 360° , hence $e(n)=360^\circ/n$ [2].

13(c). $e(12)=30^\circ$ [2].

13(d). An irregular polygon still has exterior angle sum 360° but its individual angles are not equal [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.